

Aditya Teja Vadlamani

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Research Interests

Graph Machine Learning, Conformal Prediction, High-Performance Computing, Recommender Systems, Generative AI

Education

The Ohio State University

Columbus, OH

PhD in Computer Science and Engineering

Aug 2023 – Present

Advisor: Dr. Srinivasan Parthasarathy

M.S. in Computer Science and Engineering

Jan 2022 – May 2023

Project: An Open-Source Implementation of MultiBiSage and Renewed Bearing

Advisor: Dr. Srinivasan Parthasarathy

B.S. in Computer Science and Engineering, Mathematics, *Magna Cum Laude*

Aug 2018 – May 2022

Publications and Preprints

- **Vadlamani, A. T.**, Salarian, S., Gurukar, S., Parthasarathy, S. *BLB-HGNN: Bag of Little Bootstraps for Training Heterogeneous GNNs*. IEEE International Conference on Data Mining (ICDM), November 2025.
- **Vadlamani, A. T.***, Maneriker, P.*, Srinivasan, A., He, Y., Payani, A., Parthasarathy, S. *Conformal Prediction: A Theoretical Note and Benchmarking Transductive Node Classification in Graphs*. Transactions on Machine Learning Research (TMLR), May 2025
- **Vadlamani, A. T.***, Srinivasan, A.*, Maneriker, P., Payani, A., Parthasarathy, S. *A Generic Framework for Conformal Fairness*. The Thirteenth International Conference on Learning Representations (ICLR), April 2025.
 - This work was also accepted for the Responsible AI Day at KDD 2025
- Dias, V., Ferraz, S., **Vadlamani, A. T.**, Erfanian, M., Teixeira, C. H. C., Guedes, D., Meira Jr, W., Parthasarathy, S. *Graph Pattern Mining Paradigms: Consolidation and Renewed Bearing*. IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC), December 2023. Presenting Author. Best Paper Nominee.
- **Vadlamani, A. T.**, Rumreich, L., Phillips, A. *GIFTS: Exploration Activities for Just-in-Time Learning in a First-Year Engineering Robotics Design-Build Project*. Proceedings from the American Society for Engineering Education (ASEE) Annual Conference, June 2023

Under Review

- **Vadlamani, A. T.***, Srinivasan, A.*, Meghraz, A., Parthasarathy, S. *FedCF: Fair Federated Conformal*.

* Denotes shared first authorship.

Honors and Awards

- Recipient of the **University Fellowship** from the Graduate School of The Ohio State University. 2023-2024
- Best Paper Nominee at HiPC 2023 (top 6 out of 36 regular papers) December 2023

Research Experience

The Ohio State University

Columbus, OH

Graduate Student Researcher

Aug 2022 – Present

- Multiple projects in scalable graph ML & graph mining, conformal prediction, and fairness in ML.
- Administer and manage the research group's multi-GPU high-performance compute infrastructure supporting large-scale ML and data-intensive research.

MS Research Project

Aug 2022 – May 2023

- Reimplemented a proprietary heterogeneous GNN using PyTorch and Ray to assess reproducibility and explored extensions via sampling and ensembling techniques. This work led to an ICDM publication.

Industry Experience

Google

Sunnyvale, CA

PhD Software Development Engineer Intern

May 2025 – Aug 2025

- Designed and implemented a GNN-based model to rank which code tests in the Google codebase are most likely to break when a change is made to an ML library.

Cruise

Sunnyvale, CA

PhD Software Development Engineer Intern

May 2023 – Aug 2023

- Designed and implemented a prototype encoder model using PyTorch Lightning to detect invalid simulated scenarios of the autonomous vehicle, reducing the number of manual reviews and time for software releases by several days.

Amazon Web Services

Cupertino, CA

Software Development Engineer Intern

May 2022 – Aug 2022

- Prototyped a scalable AWS environment for functional network verification of routing policies.
- Implemented syntactic validation for the routing policy attributes datastore to prevent numeric and regex issues, eliminating unnecessary redeployments to routers, which can take weeks to months.

Amazon

Virtual

Software Development Engineer Intern

May 2021 – Aug 2021

- Designed, built, and deployed a serverless full-stack single-page application that engineers use to record ground-truth information for the team's search service.

CAS

Columbus, OH

Technology Intern

May 2019 – Aug 2019

- Developed the front end for a full-stack account recovery application to enhance customer experience and reduce customer support call times by up to 15 minutes.

Teaching Experience

The Ohio State University

Columbus, OH

Graduate Teaching Associate

Jan 2022 – May 2023; Aug 2024 – Present

- Taught the laboratory and (robot) project components for the honors first-year engineering sequence.
- Led software development efforts, including development of program-wide web applications and communication software for Arduino, Raspberry Pi, and custom PCB components for automated scoring and aesthetic LEDs.
- Fostered teamwork and developed the obstacle course for the first-year robot project with a multidisciplinary group.

Projects

- Online Store** – Developed a full-stack application using Vue.js and Express.js and hosted on AWS via Terraform. The application serves over 2,000 students and 100 instructional staff of the first-year engineering programs at OSU.
- Movie Recommendation Web App** – Built a movie recommendation app using Flask, React.js, and Elasticsearch.
- The Legend of Zelda: Dungeon 1** – Recreated the first dungeon of *The Legend of Zelda SNES* game using Visual C# and the Monogame game engine with a team of developers while following Agile practices.
- Robot Design Project** – Designed, built, and coded an autonomous robot with a team of engineers to navigate a course and complete several tasks. Was awarded [2nd place](#) for *Outstanding Achievement in Innovation*.

Skills

Programming Languages: Python, C++, Java, SQL, MATLAB, JavaScript

Machine Learning & Data Science: PyTorch, TensorFlow, NumPy, Pandas, scikit-learn

Graph Machine Learning: DGL (Deep Graph Library), PyTorch Geometric, TF-GNN, NetworkX

Big Data & Parallel Computing: Apache Spark, Apache Flume, Ray, SLURM, CUDA

Cloud Platforms & DevOps: AWS, Google Cloud Platform, Terraform, Git

Web Development & Frameworks: Vue.js, React.js, Flask, Express.js